

Scalability & Future Plan

Date	15 March 2026
Team ID	SWTID-2026-4683
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current EduGenie solution can be scaled to handle larger user bases, increased loads, and extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dependence on external Gemini API availability and quotas	API delays or limits can affect response availability.	High
2	Limited persistent user data and learning-history storage in the current prototype	Less personalized long-term learning experience.	Medium
3	Local/development-oriented deployment	Scaling to many concurrent users requires production hosting and resource management.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Prototype suitable for small-scale use.	Use scalable cloud hosting with load balancing and autoscaling.
2	Data Storage	Limited persistent storage in the prototype.	Add a managed database for profiles, learning history, quiz results, and recommendations.
3	Performance	Backend handles Gemini API requests.	Add caching where appropriate, asynchronous processing, request optimization, monitoring, and rate-limit handling.
4	Security	API credentials use environment configuration.	Use secure secret management, authentication, HTTPS, input validation, logging, and access controls.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Production deployment, improved error handling, monitoring, and API quota management	0–3 months	Improved reliability and smoother user experience.
Phase 2	User accounts, learning history, progress tracking, and persistent quiz results	3–6 months	Personalized and continuous learning experience.
Phase 3	Adaptive recommendations, richer analytics, and expanded quiz/question generation	6–9 months	More effective personalization and measurable learning progress.
Phase 4	Mobile/PWA support, multilingual learning, advanced AI features, and larger-scale cloud deployment	9–12+ months	Broader accessibility and support for a larger user base.

Project: EduGenie: Google Gemini Powered Learning Assistant **Name:** Dechen